

CBF-Based Hierarchical Quadratic Programs with Guaranteed Feasibility for Safety-Critical Systems

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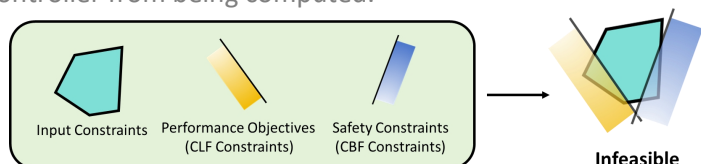
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Motivation

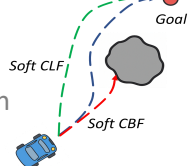
- Infeasibility** under multiple hard constraints

In CLF-CBF-QP-based controller design frameworks, multiple constraints may conflict with each other, making the optimization problem infeasible and preventing the desired controller from being computed.



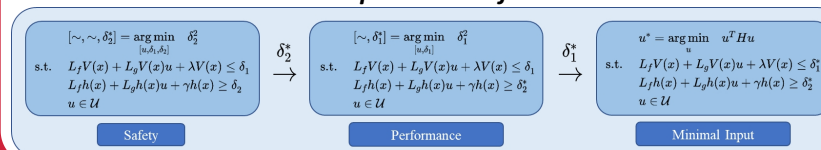
- Performance degradation** under weighted soft constraints

Soft constraints can ensure feasibility, but may degrade the performance guaranteed by the constraints, even when the system can strictly satisfy the prescribed requirements.



Safety-first CLF-CBF QP

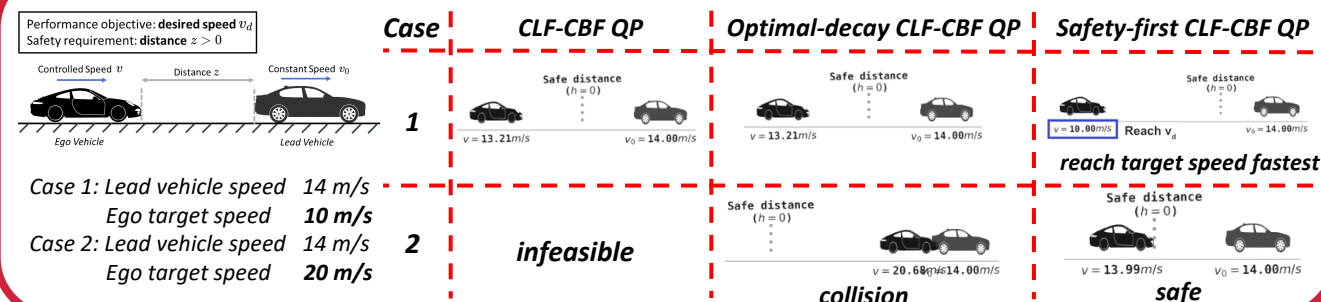
A hierarchical optimization framework



Core idea:

compute the minimum relaxation sequentially according to constraint priorities.

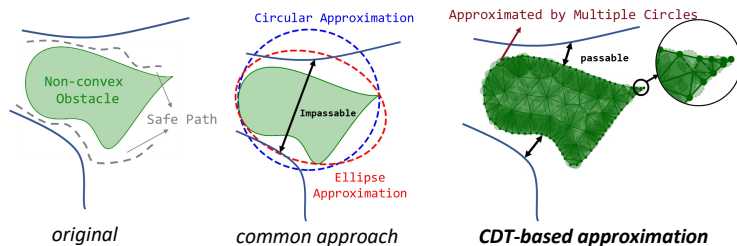
Case Study: Adaptive Cruise Control



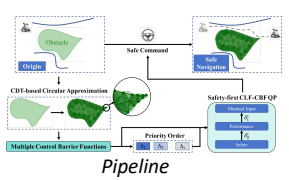
Safe Navigation

Constrained Delaunay Triangulation (CDT)-based Approximation

Our method can handle multiple constraints. Therefore, we approximate irregular obstacles with multiple circles to improve performance in narrow and cluttered environments.



Real-world Experiments



# of circles	1	5	10	20
CLF-CBF QP	1.7ms	2.1ms	2.4ms	5.4ms
Optimal-decay CLF-CBF QP	1.9ms	2.2ms	2.5ms	5.8ms
Safety-first CLF-CBF QP	3.7ms	7.5ms	12.9ms	23.5ms

*Safety-first CLF-CBF QP remains real-time with multiple CBF constraints (~80 Hz with 10 circular CBFs).

